

# Mathematical Foundations of Political Methodology

Robert Gulotty

Associate Professor  
Department of Political Science  
University of Chicago

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- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

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# Points and Vectors

## Definition

*A point  $\mathbf{x} \in R^n$  is an ordered  $n$ -tuple,  $(x_1, x_2, \dots, x_n)$ . The vector  $\mathbf{x} \in R^n$  is the arrow pointing from the origin  $(0, 0, \dots, 0)$  to  $\mathbf{x}$ .*

Some Examples:

- (Latitude, Longitude, Elevation)
- (1, 2, 3)
- (0, 1, 0)

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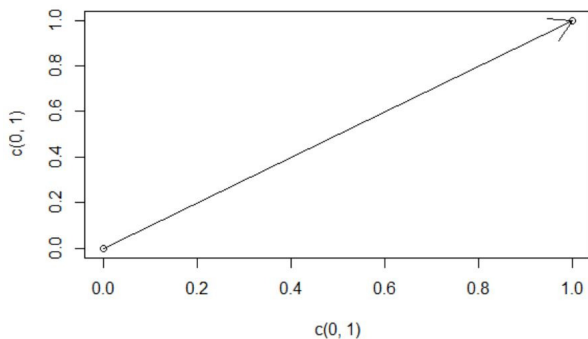
*A point  $\mathbf{x} \in R^n$  is an ordered  $n$ -tuple,  $(x_1, x_2, \dots, x_n)$ . The vector  $\mathbf{x} \in R^n$  is the arrow pointing from the origin  $(0, 0, \dots, 0)$  to  $\mathbf{x}$ .*

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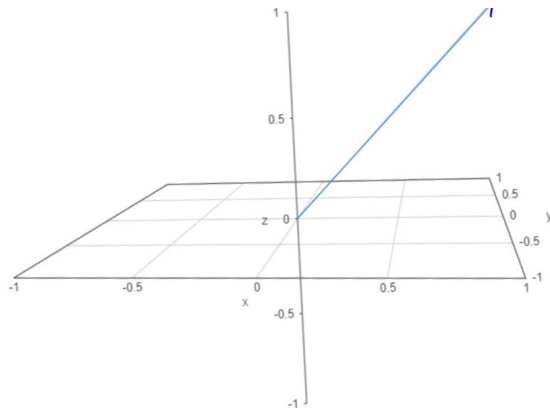
# Points and Vectors

Now, let's see a vector in a 2 dimensional space.



# Points and Vectors

Now, let's see a vector in a 3 dimensional space.





# Arithmetic with Vectors

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Suppose  $\mathbf{u}$  and  $\mathbf{v}$  are vectors  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ ,

$$\mathbf{u} = (u_1, u_2, u_3, \dots, u_n)$$

$$\mathbf{v} = (v_1, v_2, v_3, \dots, v_n)$$

We will say  $\mathbf{u} = \mathbf{v}$  if  $u_1 = v_1, u_2 = v_2, \dots, u_n = v_n$

Define the sum of  $\mathbf{u} + \mathbf{v}$  as

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3, \dots, u_n + v_n)$$

Suppose  $k \in \mathbb{R}$ . We will call  $k$  a scalar.

Define  $k\mathbf{u}$  as the scalar product

$$k\mathbf{u} = (ku_1, ku_2, \dots, ku_n)$$

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# Examples

Suppose:

$$\mathbf{u} = (1, 2, 3, 4, 5)$$

$$\mathbf{v} = (1, 1, 1, 1, 1)$$

$$k = 2$$

Then,

$$\mathbf{u} + \mathbf{v} = (1 + 1, 2 + 1, 3 + 1, 4 + 1, 5 + 1) = (2, 3, 4, 5, 6)$$

$$k\mathbf{u} = (2 \times 1, 2 \times 2, 2 \times 3, 2 \times 4, 2 \times 5) = (2, 4, 6, 8, 10)$$

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# Euclidean Norm of a Vector

Now, a question will be: how do we measure the length of a vector.

One way is to use Euclidean norm to measure the length of a vector

## Definition

*On an  $n$ -dimensional Euclidean space  $R^n$ , the Euclidean norm of vector,  $x = (x_1, x_2, \dots, x_n)$ , is defined by the formula*

$$\|x\|_2 := \sqrt{x_1^2 + \dots + x_n^2}.$$

An example. What's the Euclidean norm of  $x = (1, 3, 5)$ ?

Its Euclidean norm equals to:

$$\sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$$



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- Vectors
- **Matrix**
- Matrix Algebra
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# Matrices

Now, let's see some examples for using matrices.

The first example is the following system of equations:

$$2x + 2y + 3z = 1$$

$$3x + 5y + 7z = 2$$

$$5x + 8y + 3z = 3$$

The coefficient of unknowns can be written compactly as matrices:

$$\begin{bmatrix} 2 & 2 & 3 \\ 3 & 5 & 7 \\ 5 & 8 & 3 \end{bmatrix}$$

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# Matrices

A second example of matrix is the data set.

|    | id | name                    | year | urbanpop | totalpop | urbrate  | fpresence | westelbe | protestant | latitude | longitude |
|----|----|-------------------------|------|----------|----------|----------|-----------|----------|------------|----------|-----------|
| 1  | 1  | Rhineland               | 1700 | 83000    | .        | .        | 19        | 1        | .25        | 50.35    | 7.6       |
| 2  | 1  | Rhineland               | 1750 | 129000   | 1438817  | 8.9657   | 19        | 1        | .25        | 50.35    | 7.6       |
| 3  | 1  | Rhineland               | 1800 | 274000   | 1860719  | 14.72549 | 19        | 1        | .25        | 50.35    | 7.6       |
| 4  | 1  | Rhineland               | 1850 | 572278   | 2846708  | 20.10317 | 19        | 1        | .25        | 50.35    | 7.6       |
| 5  | 1  | Rhineland               | 1875 | 1629051  | 3804381  | 42.8204  | 19        | 1        | .25        | 50.35    | 7.6       |
| 6  | 1  | Rhineland               | 1880 | 1850627  | 4074000  | 45.42531 | 19        | 1        | .25        | 50.35    | 7.6       |
| 7  | 1  | Rhineland               | 1885 | 2103753  | 4344527  | 48.42306 | 19        | 1        | .25        | 50.35    | 7.6       |
| 8  | 1  | Rhineland               | 1895 | 2836140  | 5106002  | 55.54522 | 19        | 1        | .25        | 50.35    | 7.6       |
| 9  | 1  | Rhineland               | 1900 | 3506587  | 5759798  | 60.88038 | 19        | 1        | .25        | 50.35    | 7.6       |
| 10 | 1  | Rhineland               | 1905 | 4145633  | 6436337  | 64.40982 | 19        | 1        | .25        | 50.35    | 7.6       |
| 11 | 1  | Rhineland               | 1910 | 4806725  | 7121140  | 67.49937 | 19        | 1        | .25        | 50.35    | 7.6       |
| 12 | 2  | Mark                    | 1700 | 5000     | 117963   | 4.238618 | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 13 | 2  | Mark                    | 1750 | 5000     | 149869   | 3.336254 | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 14 | 2  | Mark                    | 1800 | 36000    | 193096   | 18.64353 | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 15 | 2  | Mark                    | 1850 | 55628    | 362392   | 15.35011 | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 16 | 2  | Mark                    | 1875 | 305301   | 721466   | 42.31675 | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 17 | 2  | Mark                    | 1880 | .        | .        | .        | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 18 | 2  | Mark                    | 1885 | .        | .        | .        | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 19 | 2  | Mark                    | 1895 | .        | .        | .        | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 20 | 2  | Mark                    | 1900 | 1134655  | 1514546  | 74.91717 | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 21 | 2  | Mark                    | 1905 | .        | .        | .        | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 22 | 2  | Mark                    | 1910 | .        | .        | .        | 6         | 1        | .6         | 51.68333 | 7.816667  |
| 23 | 3  | Westphalia without Mark | 1700 | 7000     | .        | .        | 6         | 1        | .4         | 51.96667 | 7.633333  |
| 24 | 3  | Westphalia without Mark | 1750 | 14000    | 528851   | 2.647246 | 6         | 1        | .4         | 51.96667 | 7.633333  |
| 25 | 3  | Westphalia without Mark | 1800 | 55000    | 763110   | 7.20735  | 6         | 1        | .4         | 51.96667 | 7.633333  |
| 26 | 3  | Westphalia without Mark | 1850 | 61000    | 1120946  | 5.441833 | 6         | 1        | .4         | 51.96667 | 7.633333  |
| 27 | 3  | Westphalia without Mark | 1875 | 190503   | 1184231  | 16.08664 | 6         | 1        | .4         | 51.96667 | 7.633333  |
| 28 | 3  | Westphalia without Mark | 1880 | .        | .        | .        | 6         | 1        | .4         | 51.96667 | 7.633333  |

# Matrices

## Definition

*A Matrix is a rectangular array of numbers*

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

*If  $\mathbf{A}$  has  $m$  rows  $n$  columns (Roman Catholic; Rum & Cake) we will say that  $\mathbf{A}$  is an  $m \times n$  matrix.*

*Suppose  $\mathbf{X}$  and  $\mathbf{Y}$  are  $m \times n$  matrices. Then  $\mathbf{X} = \mathbf{Y}$  if  $x_{ij} = y_{ij}$  for all  $i$  and  $j$*

## Simple Examples

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

If  $\mathbf{I}$  is an  $n \times n$  matrix we will call an identity matrix.

## Simple Examples

$$\mathbf{X} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \end{pmatrix}$$

$\mathbf{X}$  is an  $2 \times 3$  matrix

- Vectors
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# Matrix Algebra

## Definition

Suppose  $\mathbf{X}$  and  $\mathbf{Y}$  are  $m \times n$  matrices. Then define

$$\begin{aligned}\mathbf{X} + \mathbf{Y} &= \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} + \begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{m1} & y_{m2} & \cdots & y_{mn} \end{pmatrix} \\ &= \begin{pmatrix} x_{11} + y_{11} & x_{12} + y_{12} & \cdots & x_{1n} + y_{1n} \\ x_{21} + y_{21} & x_{22} + y_{22} & \cdots & x_{2n} + y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} + y_{m1} & x_{m2} + y_{m2} & \cdots & x_{mn} + y_{mn} \end{pmatrix}\end{aligned}$$

# Matrix Algebra

## Definition

Suppose  $\mathbf{X}$  is an  $m \times n$  matrix and  $k \in R$ . Then,

$$k\mathbf{X} = \begin{pmatrix} kx_{11} & kx_{12} & \dots & kx_{1n} \\ kx_{21} & kx_{22} & \dots & kx_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ kx_{m1} & kx_{m2} & \dots & kx_{mn} \end{pmatrix}$$

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$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \color{red}{x_{21}} & \color{red}{x_{22}} & \dots & \color{red}{x_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$
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If  $\mathbf{X}$  is an  $m \times n$  then  $\mathbf{X}'$  is  $n \times m$ .

If  $\mathbf{X} = \mathbf{X}'$  then we say  $\mathbf{X}$  is symmetric.

# Matrix Transpose

Example 1:  $\mathbf{X} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$  then  $\mathbf{X}' = \begin{pmatrix} 4 & 1 \\ 1 & 2 \\ 2 & 3 \end{pmatrix}$



# Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

We will create a new matrix  $\mathbf{A}$  by matrix multiplication:

$$\begin{aligned} \mathbf{A} &= \mathbf{XY} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \end{aligned}$$

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How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

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# Matrix Multiplication

## Definition

Suppose  $\mathbf{X}$  is an  $m \times n$  matrix and  $\mathbf{Y}$  is an  $n \times k$  matrix. Then define the matrix  $\mathbf{A}$  an  $m \times k$  matrix that obtains from multiplying  $\mathbf{X}$  and  $\mathbf{Y}$  as,

$$\begin{aligned}\mathbf{A} &= \mathbf{XY} \\ &= \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} \begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1k} \\ y_{21} & y_{22} & \cdots & y_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nk} \end{pmatrix} \\ &= \begin{pmatrix} x_{11}y_{11} + x_{12}y_{21} + \cdots + x_{1n}y_{n1} & \cdots & x_{11}y_{1k} + x_{12}y_{2k} + \cdots + x_{1n}y_{nk} \\ \vdots & \ddots & \vdots \\ x_{m1}y_{11} + x_{m2}y_{21} + \cdots + x_{mn}y_{n1} & \cdots & x_{m1}y_{1k} + x_{m2}y_{2k} + \cdots + x_{mn}y_{nk} \end{pmatrix}\end{aligned}$$

## Definition

Suppose  $\mathbf{X}$  is an  $m \times n$  matrix and  $\mathbf{Y}$  is an  $n \times k$  matrix. Write the row

vectors of  $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_m \end{pmatrix}$  and  $\mathbf{Y}$  as column vector  $\mathbf{Y} = (\mathbf{y}_1 \ \mathbf{y}_2 \ \dots \ \mathbf{y}_k)$ .

Then the  $m \times k$  matrix  $\mathbf{A} = \mathbf{X}\mathbf{Y}$  can be written as

$$\mathbf{A} = \begin{pmatrix} \mathbf{x}_1 \cdot \mathbf{y}_1 & \mathbf{x}_1 \cdot \mathbf{y}_2 & \dots & \mathbf{x}_1 \cdot \mathbf{y}_k \\ \mathbf{x}_2 \cdot \mathbf{y}_1 & \mathbf{x}_2 \cdot \mathbf{y}_2 & \dots & \mathbf{x}_2 \cdot \mathbf{y}_k \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_m \cdot \mathbf{y}_1 & \mathbf{x}_m \cdot \mathbf{y}_2 & \dots & \mathbf{x}_m \cdot \mathbf{y}_k \end{pmatrix}$$

# Matrix Multiplication

Let's work on an example together!

$$\mathbf{X} = \begin{pmatrix} 1 & 4 & 5 \\ 10 & 2 & 3 \end{pmatrix}$$

$$\mathbf{Y} = \begin{pmatrix} 2 & 3 \\ 1 & 5 \\ 3 & 5 \end{pmatrix} \text{ What is } \mathbf{XY}?$$

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Not all matrices can be multiplied.

Matrix  $\mathbf{AB}$  exists only if the number of columns in  $\mathbf{A}$  = number of rows in  $\mathbf{B}$ . If  $\mathbf{AB}$  exists we will say the matrices are conformable



# Matrix Multiplication with a Vector

Suppose  $\mathbf{X} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 1 & 5 & 1 & 2 \\ 3 & 5 & 3 & 4 \end{pmatrix}$  a  $3 \times 4$  matrix and that  $\mathbf{v} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 10 \end{pmatrix}$  a  $4 \times 1$

matrix (or a column vector) what is

$\mathbf{X}\mathbf{v}$ ?

What is  $\mathbf{X}'\mathbf{v}$ ?

# Algebraic Properties

Suppose  $\mathbf{X}$  is an  $m \times n$  matrix and  $\mathbf{Y}$  is an  $n \times k$  matrix. Suppose that

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \text{ as the identity matrix and that } k \in R.$$

- $\mathbf{XY} \neq \mathbf{YX}$  in general
- $\mathbf{XI} = \mathbf{X}$
- $(\mathbf{X}')' = \mathbf{X}$
- $(\mathbf{XY})' = \mathbf{Y}'\mathbf{X}'$
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- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

# Matrix Inversion: Motivations

Big topic: suppose  $\mathbf{X}$  is an  $n \times n$  matrix. We want to find the matrix  $\mathbf{X}^{-1}$  such that

$$\mathbf{X}^{-1}\mathbf{X} = \mathbf{X}\mathbf{X}^{-1} = \mathbf{I}$$

where  $\mathbf{I}$  is the  $n \times n$  identity matrix.

Why do we care about matrix inversion?

It will help us to determine whether a system of equations is solvable.

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Consider the following equations:

$$x_1 + x_2 + x_3 = 0$$

$$0x_1 + 5x_2 + 0x_3 = 5$$

$$0x_1 + 0x_2 + 3x_3 = 6$$

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\mathbf{x} = (x_1, x_2, x_3)$$

$$\mathbf{b} = (0, 5, 6)$$

The system of equations are now,

$$\mathbf{Ax} = \mathbf{b}$$

$\mathbf{A}^{-1}$  exists if and only if  $\mathbf{Ax} = \mathbf{b}$  has only one solution.

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# Matrix Inversion, Definition

## Definition

*Suppose  $\mathbf{X}$  is an  $n \times n$  matrix. We will call  $\mathbf{X}^{-1}$  the inverse of  $\mathbf{X}$  if*

$$\mathbf{X}^{-1}\mathbf{X} = \mathbf{X}\mathbf{X}^{-1} = \mathbf{I}$$

*If  $\mathbf{X}^{-1}$  exists then  $\mathbf{X}$  is invertible. If  $\mathbf{X}^{-1}$  does not exist, then we will say  $\mathbf{X}$  is singular.*

# Matrix Inversion

Now, let's discuss some properties of matrix inversion.

## Theorem

*The inverse of matrix  $X$ ,  $X^{-1}$ , is unique*

Proof.

By way of contradiction, suppose not. Then there are at least two matrices  $A$  and  $C$  such that  $AX = I$  and  $CX = I$

This implies that,

$$\begin{aligned}AXC &= (AX)C \\ &= IC \\ &= C\end{aligned}$$

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# Matrix Inversion

But it also implies that

$$\begin{aligned} \mathbf{A}\mathbf{X}\mathbf{C} &= \mathbf{A}(\mathbf{X}\mathbf{C}) \\ &= \mathbf{A}(\mathbf{I}) \\ &= \mathbf{A} \end{aligned}$$

So  $\mathbf{C} = \mathbf{A}\mathbf{X}\mathbf{C} = \mathbf{A}$  or  $\mathbf{C} = \mathbf{A}$  but this contradicts our assumption that there are two unique inverses.

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# Matrix Inversion

## Theorem

Suppose  $\mathbf{A}$  has inverse  $\mathbf{A}^{-1}$  and  $\mathbf{B}$  has inverse  $\mathbf{B}^{-1}$ . Then,

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

Proof.

We need to show that  $(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = (\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$ .

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# Matrix Inversion

## Theorem

Suppose  $\mathbf{A}$  has inverse  $\mathbf{A}^{-1}$  and  $\mathbf{B}$  has inverse  $\mathbf{B}^{-1}$ . Then,

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## Inclass Exercise: Challenge Inversion Proofs

- Show that  $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$ .
- Show that  $(k\mathbf{A})^{-1} = \frac{1}{k}\mathbf{A}^{-1}$

# Matrix Inversion

After we define matrix inversion and discuss its properties, a natural question to ask is that when do matrix inversion exist?

The answer relates to a concept, linear independence.

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# Matrix Inversion: Existence

## Definition

*Suppose we have a set of vectors  $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$*

*And consider the system of equations*

$$k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \dots + k_r \mathbf{v}_r = \mathbf{0}$$

*If the only solution is  $k_1 = 0, k_2 = 0, k_3 = 0, \dots, k_r = 0$  then we say that the set is linearly independent. If there are other solutions, then the set is linearly dependent.*

# Matrix Inversion: Existence

Consider  $\mathbf{v}_1 = (1, 0, 0)$ ,  $\mathbf{v}_2 = (0, 1, 0)$ ,  $\mathbf{v}_3 = (0, 0, 1)$

Can we write this as a combination of other vectors?

Consider  $\mathbf{v}_1 = (1, 0, 0)$ ,  $\mathbf{v}_2 = (0, 1, 0)$ ,  $\mathbf{v}_3 = (0, 0, 1)$ ,  $\mathbf{v}_4 = (1, 2, 3)$ .

Can we write any of these vector as a combination of the remaining two vectors?

$$\mathbf{v}_4 = \mathbf{v}_1 + 2\mathbf{v}_2 + 3\mathbf{v}_3$$

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*Suppose  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_K \in \mathbb{R}^n$ . If  $K > n$  then the set is linearly dependent*

- $\mathbf{v}_1 = (v_{11}, v_{21}, \dots, v_{n1})$
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Suppose  $\mathbf{X}$  is an  $n \times n$  matrix. Recall we can write this matrix as  $\begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{pmatrix}$ .

Then  $\mathbf{X}$  has an inverse if and only if  $S = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$  is linearly independent

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# Matrix Inversion: Rank and Span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3; the rank of  $\mathbf{A} = 2$

These three vectors span  $\mathbb{R}^2$  but not  $\mathbb{R}^3$ : Show that for any  $(x_1, x_2) \in \mathbb{R}^2$ , we can find a  $\lambda_1, \lambda_2, \lambda_3$  such that

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## Finding the rank of a matrix $\mathbf{A}$

- Reduce  $\mathbf{A}$  to an echelon matrix  $\mathbf{E}$  using Gaussian elimination
- The number of non-zero rows of  $\mathbf{E}$  is the rank of  $\mathbf{A}$ !

## Properties of ranks

- The rank of an  $n \times m$  matrix  $\mathbf{A}$  can't be greater than either  $n$  or  $m$ :  
 $\text{Rank}(\mathbf{A}) \leq \min(n, m)$
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Take

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Gauss-Jordan elimination is similar to Gaussian elimination, but turns matrix into a diagonal matrix, instead of just an echelon matrix

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$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

Step 1:  $\begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$

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Since we are creating a diagonal we want to get rid of the “1” in (1, 2)

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This added  $2 \times$  Row 2 to Row 1 and subtracted Row 2 from Row 3

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## Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Since we are creating a diagonal we want to get rid of the “1” in (1, 2)

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

This added  $2 \times \text{Row 2}$  to Row 1 and subtracted Row 2 from Row 3

## Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding  $2 \times \text{Row 3}$  to Rows 1 & 2

$$\text{Step 4: } \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$



## Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding  $2 \times \text{Row 3}$  to Rows 1 & 2

$$\text{Step 4: } \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

## Matrix Inversion: Gauss-Jordan elimination

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## Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding  $2 \times \text{Row 3}$  to Rows 1 & 2

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We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

# Matrix Inversion: An Application

Suppose we have following system of equations:

$$2x + 1y + 2z = 1$$

$$3x + 1y + 1z = 3$$

$$3x + 1y + 2z = 5$$

Say,  $\mathbf{x} = (x, y, z)'$ ,  $\mathbf{b} = (1, 3, 5)'$ .

Say, the coefficient matrix for unknowns is  $A$ , the inverse of  $A$  is computed by the previous example.

Then, we can solve the unknowns by:

$$\mathbf{x} = A^{-1}\mathbf{b} = (4, -11, 2)'$$